

Project Title: Stock Price Analysis and Forecasting using Time Series Data

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Abstract:-

This project focuses on analyzing historical stock price data using exploratory data analysis and time-series forecasting techniques. The dataset was cleaned and processed to make it analysis-ready. Visualizations were used to understand trends, volatility, and return behavior. Baseline forecasting models such as Naïve Forecast and Moving Average were implemented to predict future prices. The project demonstrates the complete data science workflow from data preprocessing to forecasting.

Introduction:-

Stock markets generate a large amount of data every day in the form of stock prices.

This data is recorded over time and is known as time-series data, where each record represents the price of a stock on a particular date.

Stock price data usually includes values such as Open, High, Low, Close, and Volume, which help in understanding how a stock behaves in the market. Analyzing this historical data is important because stock prices often follow patterns, trends, and seasonal movements.

Stock price forecasting is important for investors, analysts, and financial institutions because it helps them

make better decisions. By studying past price movements, investors can identify trends, understand market behavior, and reduce risk while planning future investments.

Dataset Description:-

The dataset used in this project contains historical daily stock price data. It represents how the stock price changes over time and is suitable for time-series analysis and forecasting.

📌 Source of Data

The dataset was obtained from an online financial data source (such as Yahoo Finance) and stored in CSV format. It contains real-world stock market data used for analysis and learning purposes.

📌 Time Range

The dataset covers daily stock prices over a specific time period (for example, from 2023 to 2024).

Each row in the dataset represents the stock price information for one trading day.

📌 Columns Description

The dataset includes the following columns:

Date - The trading date of the stock (used as the time index)

Open - Price at which the stock opened on that day

High - Highest price reached during the day

Low - Lowest price reached during the day

Close - Closing price of the stock (used as the target variable)

Volume - Total number of shares traded on that day

📌 Data Type

The data is time-series data

Frequency: Daily

Index: Date column

📌 Purpose of the Dataset

This dataset is used to:

Analyze historical stock price behavior

Identify trends and patterns through visualization

Apply basic time-series forecasting techniques to predict future prices

Data Cleaning & Preprocessing:-

Include:

1. Renaming columns

2. Removing unwanted rows

3. Fixing data types

4. Handling missing values

5. Sorting by date

The raw dataset contained non-data rows and incorrect column names. These issues were resolved by renaming columns, removing invalid rows, converting date columns to datetime format, and ensuring numerical consistency. The cleaned dataset was saved separately for further analysis.

Exploratory Data Analysis (EDA):-

Include:

1. Line plots

2. Return analysis

3. Volatility

4. Distribution analysis

5. Pie charts (positive vs negative days)

EDA was performed to understand price movement, daily returns, and volatility. Visualizations using Matplotlib and Seaborn helped identify trends and fluctuations in stock prices. Pie charts were used to analyze the proportion of positive and negative return days.

Visualization Techniques:-

Data visualization was carried out using Matplotlib and Seaborn to clearly represent trends and patterns in the data. Line charts, distribution plots, and pie charts were used to support analytical insights.

Forecasting Methodology:-

Include models:

1.Naive Forecast

2.Moving Average Forecast

Baseline forecasting models were implemented to predict future stock prices. The Naive Forecast assumes that future prices will be equal to the last observed value, while the Moving Average Forecast uses the average of past observations. These models serve as reference benchmarks rather than trend-learning models.

Results & Discussion:-

The forecasts produced flat trends because baseline models do not learn temporal patterns. They are designed to provide reference performance rather than capture complex trends.

Conclusion:-

This project successfully demonstrated the complete workflow of time-series data analysis and forecasting. Although simple baseline models were used, the project provided strong foundational understanding of forecasting concepts and data preparation techniques.